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Gimmick or genuineness? Exploring the antecedents of AI virtual streamers aversion in live-streaming commerce

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ABSTRACT

AI virtual streamers are emerging as a novel presence in live-streaming commerce. However, a considerable portion of consumers are still averse to these AI entities. While existing research predominantly emphasizes the positive impacts of AI virtual streamers, it often overlooks the factors contributing to consumer aversion. To address this critical yet underexplored gap, this study employs the antecedent-belief-consequence (ABC) framework to examine the determinants of consumer aversion to AI virtual streamers. Data collected from 402 consumers with a comprehensive understanding of AI virtual streamers was conducted using the partial least squares-structural equation modeling (PLS-SEM) approach. The results validate the impact of two key antecedents (i.e., anthropomorphism and technophobia) on the consequences (AI virtual streamers aversion) and explore the underlying mechanism of beliefs through the reflexive stage (i.e., perceived unwarm and perceived incompetent) and the reflective stage (i.e., consumer resonance and disfluency). Additionally, multigroup analysis (MGA) based on self-construal indicates significant group differences, with interdependent (independent) consumers being more likely to exhibit aversion through the perceived unwarm (perceived incompetent) pathway. These findings contribute to the current body of research by clarifying the multi-stage and multi-dimensional processes through which technophobia and anthropomorphism drive AI virtual streamers aversion. Practitioners can leverage these insights to address the underlying causes of consumer aversion, thereby facilitating the integration of AI virtual streamers into live-streaming commerce.

1. Introduction

The advancement of artificial intelligence (AI) technology and the expansion of the metaverse have made AI virtual streamers a popular choice for live-streaming commerce platforms and merchants seeking to provide real-time customer service (Xiao et al., 2025). These AI virtual streamers have become a fresh attraction, such as YouTube Live, where they initially gained popularity. The market for AI virtual streamers is rapidly expanding, with the global Virtual YouTuber market valued at USD 2885.9 million in 2023 and projected to reach an estimated scale of USD 22,410 million by 2030 (Linkedln, 2024). These statistics underscore the effectiveness and growing appeal of employing AI virtual streamers to capture the attention of global consumers and drive sales.

While AI virtual streamers offer businesses new opportunities to engage consumers and enhance profitability, a considerable portion of consumers are averse to them. According to a report by JIEMIAN News, some merchants have observed that AI virtual streamers fail to achieve anticipated conversion rates and elicit increased consumer complaints due to their inflexible responses and inability to address inquiries effectively (Jiemian, 2024). This finding is further supported by a comprehensive 2024 survey conducted by iiMedia (iiMedia, 2024), which indicated that approximately 45 % of Chinese respondents expressed reservations about AI virtual streamers in live commerce, with 8 % holding a negative perception. The deployment of AI virtual streamers is still in its nascent stages, and a significant disparity exists between human and AI virtual streamers, with the latter yet to achieve the same level of impact (Zhao, 2022). Moreover, distinctive attributes of AI virtual streamers may evoke negative consumer perceptions and stereotypes (Chen, 2023). These negative perceptions and stereotypes can lead to unfavorable outcomes, such as emotional hostility, aversion, and resistance behaviors (Bennett and Vijaygopal, 2018). In other words, for service providers and operators, gaining deeper insights into consumer aversion toward AI virtual streamers is crucial, as such attitudes are directly linked to the success or failure of AI virtual streamers,

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which, in turn, affect profitability. Therefore, it is essential to thoroughly analyze the potential antecedents of consumer perceptions or responses to AI virtual streamers, particularly from the perspective of aversion.

In addition to the phenomena initially capturing our attention, three primary research motivations drove our exploration of AI virtual streamers aversion (see Appendix Table A). First, from a consumer behavior perspective, distinguishing between appreciation and aversion to new technologies is crucial (Mahmud et al., 2023). While studies often focus on positive outcomes like purchase intention (Hu and Ma, 2023), trust (Wang et al., 2024), and consumption willingness (Yu et al., 2023), neglecting negative consumer responses to AI virtual streamers. To address this gap, it is essential to systematically analyze the factors contributing to AI virtual streamers aversion, responding to calls for a more comprehensive investigation of AI technology (Yang et al., 2023).

Second, consumer traits are essential drivers of consumer behavioral outcomes (Agogo and Hess, 2018). However, existing research often overlooks their significance, primarily focusing on technical features such as anthropomorphism (Chen et al., 2024a, 2024b; Sun et al., 2024). Anthropomorphism and technophobia are critical antecedents that profoundly influence consumer attitudes from the perspectives of technical features and consumer traits (Sun et al., 2024; Deng and Jiang, 2023). Anthropomorphism has been found to promote trust and purchase intention (Chen et al., 2024a, 2024b; Sun et al., 2024), but it may also raise concerns about privacy (Sun et al., 2023). Most current studies have mainly concentrated on the positive effects of anthropomorphism. Our study emphasizes the underexplored negative impacts, shedding light on the overlooked adverse effects. Concurrently, technophobia is a reflection of individuals' fear regarding technology. Research by Querci et al. (2024) highlights consumers' negative responses to smart objects due to prevalent fears. In the context of AI virtual streamers, this technophobia may provoke consumer aversion. Therefore, integrating technical features (i.e., anthropomorphism) with consumer traits (i.e., technophobia) is essential for a comprehensive understanding of consumer aversion to AI virtual streamers.

Third, current research has identified the critical mediators of aversion, such as perceived eeriness and perceived affinity (Liu et al., 2024), perceived flexibility (Chang and Wang, 2023). Additionally, it has begun to explore boundary conditions in specific contexts such as technology readiness (Mahmud et al., 2023) and socio-demographic information (Liu et al., 2023); however, a deeper understanding of psychological perceptions and the influence of individual differences remains insufficient. Insight into these factors is crucial for enhancing consumer aversion behavior and developing effective marketing strategies. Consequently, this study puts forward three research questions to tackle the identified research gap:

RQ1: What factors are antecedents to consumer aversion to AI virtual streamers?

RQ2: How do these antecedents play a role in framing consumer aversion to AI virtual streamers?

RQ3: Which boundary conditions explain the relationship between these antecedents and consumer aversion to AI virtual streamers?

To answer the abovementioned questions, this research employs the antecedent-belief-consequence (ABC) framework to examine consumer aversion to AI virtual streamers. Drawing on the complexity theory, two dimensions of computers as social actors (CASA) cues (i.e., anthropomorphism) and consumer technology cues (i.e., technophobia) are identified as antecedents (A). These antecedents lead consumers to engage in various belief activities (B), including the reflexive stage and the reflective stage, which in turn trigger the consequences of AI virtual streamers aversion (C). Specifically, founded on the stereotype content model, the two aspects of negative stereotypes, i.e., perceived unwarm and perceived incompetent, were identified as the reflexive stage in the belief. Negative stereotypes play a substantial role in shaping consumer resonance and disfluency during the reflective stage, as informed by emotional resonance theory and information processing theory. These factors, in turn, directly influence subsequent behavioral consequences of AI virtual streamers aversion. Furthermore, this study attempts to consider the moderating influence of self-construal to examine the boundary conditions of consumer aversion.

Our study makes key contributions in several ways. First, this study pioneers the exploration of consumer aversion toward AI virtual streamers by introducing a novel research model, thereby contributing to the live-streaming commerce and algorithmic aversion literature. Second, we consider the impact of consumer technology cues and CASA cues on AI virtual streamers aversion, providing a systematic and comprehensive perspective on the antecedents that shape consumer aversion. Third, this study investigates the underlying mechanisms driving consumer aversion by integrating both the reflexive stage and the reflective stage into the "belief" component, thereby filling a gap in the literature concerning the psychological mechanisms behind consumer aversion. Fourth, this study examines consumers' self-construal as a moderating factor, identifying significant boundary conditions that differentiate the responses of independent versus interdependent consumers toward AI virtual streamers aversion. Our findings offer practical guidance for practitioners to improve the design of AI virtual streamers and tailor their implementation strategies with the preferences of target consumers.

2. Literature review

2.1. AI virtual streamers in live-streaming commerce

AI virtual streamers, AI-generated human-like characters that employ speech synthesis and animation technology (Miao et al., 2022), have been actively engaged in various live-streaming activities across social media platforms, including broadcasting news, gaming, educational content, etc. Furthermore, AI virtual streamers have greatly expanded their role by using algorithms to mimic a comprehensive spectrum of emotions and gestures, transforming live-streaming commerce marketing. These AI virtual streamers have been customized to cater to the particular preferences of specific consumer groups (Sands et al., 2022), offering a unique benefit to merchant brands. Compared to their human counterparts, AI virtual streamers exhibit a diminished risk of embroiling in scandals and can sustain continuous engagement with consumers, functioning without the necessity for breaks (Thomas and Fowler, 2021).

The literature on AI virtual streamers in live-streaming commerce primarily focuses on the role of AI virtual streamers in enhancing consumers' positive behavior, utilizing a variety of research methodologies, including descriptive analyses (Chen, 2023), survey studies (Sun et al., 2024), and experimental designs (Chen et al., 2024a, 2024b). Relevant studies are summarized in Appendix Table B. Research has indicated that technological features of AI virtual streamers (such as anthropomorphism, coolness, sensory language, and physical attractiveness) (Sun et al., 2024; Hu and Ma, 2023; Zhang et al., 2023; Yu et al., 2023), consumer perceptions (such as perceived ease of use, empathy, and trust) (Wan and Jiang, 2023; Xie et al., 2024; Wang et al., 2024), the type of product (Chen et al., 2024a, 2024b) can significantly enhance consumers' positive responses (such as willingness to accept; impulse buying intentions, online engagement, and purchase intention) (Chen, 2023; Zhang et al., 2023; Liu et al., 2021; Zhou et al., 2023).

Negative reactions have been recognized as significant within the information technology sector (Ali et al., 2016; Lapointe and Rivard, 2005), as understanding these responses is crucial for comprehending consumer behavior and shaping consumer perceptions. However, current research on AI virtual streamers primarily emphasizes positive aspects (Chen, 2023; Liu et al., 2021). Moreover, the literature has also found that consumer aversion fundamentally differs from appreciation behavior, implying that the factors driving aversion may not align with those that promote appreciation (Lapointe and Rivard, 2005). Therefore, there is an urgent need for a theory-driven explanatory study that

provides a greater understanding of the mechanisms behind consumer aversion to AI virtual streamers. In addition, most previous studies on the subject have been conducted from a singular perspective focusing on the technological features of AI virtual streamers. For instance, Gao et al. (2023) demonstrated that the likeability, animacy, and responsiveness of AI virtual streamers have a positive effect on social presence, telepresence, and purchase intention. Similarly, Wang et al. (2024) identified that AI virtual streamers' four traits of ability, benevolence, predictability, and integrity further increased purchase intentions through trust. While these studies have investigated the technological features of AI virtual streamers, their development ultimately relies on consumer engagement. Therefore, it is essential to integrate consumer traits into the design and evaluation of AI virtual streamers. To bridge these research gaps, the current study introduces a new dimension of consumer technology cues as antecedents based on CASA cues constructs and proposes a chain model with AI virtual streamers aversion as a consequence. This model is empirically tested using cross-sectional data, providing an in-depth analysis of the antecedents that lead to consumer aversion toward AI virtual streamers.

2.2. Antecedent-belief-consequence (ABC) framework

The ABC framework is widely used in psychology to investigate and comprehend individual cognitive and behavioral patterns (Ellis, 1962). The framework assumes that the consequences (C) of individuals' actions are not directly determined by the antecedent events (A) but rather by the individuals' beliefs (B) regarding these events. In essence, the same antecedents (A) can lead to divergent consequences (C) depending on their corresponding beliefs (B) (Ellis, 1985). The versatility of the ABC framework has been evidenced in diverse fields, including organizational work groups (Cannon and Edmondson, 2001), mobile apps (Hsu and Lin, 2016), biofuels' social acceptance (Bonaiuto et al., 2024), etc., showcasing its robust explanatory power in analyzing consumer behaviors in both online and offline contexts.

Based on these previous studies, the ABC framework is also relevant and adequate for understanding the interplay between AI virtual streamers and consumer behaviors in the live-streaming commerce context for several reasons. First, the ABC framework has been employed in digital context research to examine individual behavioral responses (Hsu and Lin, 2016; Dzandu, 2023). Since AI virtual streamers exist in a digital environment, they will also be effective for current research. Second, the ABC framework provides a multidimensional perspective for understanding the mechanisms that drive consumer behavior, offering an integrated approach to identifying key driving factors (Ellis, 1985). Notably, "antecedent" includes external and internal factors, broadening the scope of factors considered. Third, the ABC model emphasizes the continuity and dynamism of consumer behavior, which aligns with our goal of understanding consumer aversion behavior.

Specifically, "antecedent" refers to an external event or situation that triggers individuals' attitudes and behaviors, manifesting in a myriad of forms. Based on complexity theory, we consider "antecedents" in terms of CASA cues and consumer technology cues. Complexity theory posits that outcomes are not determined by a single factor but rather emerge from various antecedent conditions, creating configurations that trigger specific phenomena (Woodside, 2017). This theory has been extensively applied to clarify the complex relationships in consumer behavior (Farmaki et al., 2021; Chuah et al., 2021). In our research, CASA cues (i.e., anthropomorphism) and consumer technology cues (i.e., technophobia) are identified as key "antecedents", and their configurational combinations may elucidate consumer behavior toward AI virtual streamers. For example, existing research highlights the importance of anthropomorphism in the design and interaction of AI virtual streamers (Chen et al., 2024a, 2024b), while technophobia significantly influences consumers' attitudes and behaviors toward new technologies (Osiceanu, 2015). Therefore, a nuanced analysis of the configurational combinations of anthropomorphism and technophobia may provide valuable insights, aiding in understanding consumer attitudes and behaviors.

The second component, "belief," is defined as the individuals' interpretation or evaluation of the antecedents, encompassing both positive and negative internal thoughts and attitudes. The formation of beliefs is a complex process that involves an immediate reaction to antecedents followed by a deeper phase of reflection (Van Laer et al., 2014). Williams (2009) proposed a temporal model of rejection response comprising two distinct phases: reflexive and reflective. The reflexive stage, as described by Wirth and Williams (2009), represents the immediate, instinctive reaction to rejection. This initial phase is marked by a subconscious process, which triggers negative perceptions and shifts the individual's focus toward deeper cognitive and affective evaluation, thereby leading into the reflective stage. During the reflective phase, the individual engages in a more deliberate and introspective process, potentially transforming the initial negative perceptions into a more nuanced understanding. In this study, both the reflexive and reflective stages are conceptualized as components of "belief."

The stereotype content model offers a robust theoretical foundation for understanding the reflexive stage, as it can rapidly and unconsciously influence an individual's perception of non-human entities (El Hedhli et al., 2023). This model provides two key dimensions that explain social perceptions in interpersonal interactions: warmth and competence (Fiske, 1998). Warmth captures individual traits associated with friendliness and trustworthiness, while competence denotes individual characteristics associated with skill and efficiency (Fiske et al., 2007). These dimensions align with the subconscious reactions described in the reflexive stage, where individuals may subconsciously form initial judgments based on warmth and competence when encountering AI virtual streamers. If AI virtual streamers are perceived unwarm (i.e., not friendly and trustworthy enough) or incompetent (i.e., not skilled and efficient enough), this negative stereotype can quickly trigger adverse responses. This process occurs before deeper cognitive and affective evaluations, making the stereotype content model an apt explanatory framework for the reflexive stage. Previous studies have shown that negative stereotypes can lead to the formation of negative attitudes and behaviors. For instance, Pantano (2021) employed negative stereotypes to explore the viral impact of luxury brands on social media. Thus, the two dimensions of the negative stereotype model - perceived unwarm and perceived incompetent - can effectively explain consumer aversion to AI virtual streamers. These negative reactions are precisely the key mechanisms in the reflexivity stage.

Although the reflexive stage captures consumers' immediate responses to various cues in the antecedents, they are insufficient to represent the internal states driving consumer behavior entirely. To gain a deeper understanding of consumer motivations, it is essential to consider the reflective stage, which encompasses deeper cognitive and emotional aspects that develop over time and more significantly influence future consumer behaviors. To analyze these deeper psychological processes, we draw on emotional resonance theory and information processing theory. Emotional resonance theory suggests that the emotional connection consumers establish with a media entity significantly influences their acceptance of it (McDonnell et al., 2017). In the context of AI virtual streamers, consumer resonance can be fostered through anthropomorphic features, such as facial expressions, voice modulation, and interactive behaviors that mirror human characteristics (Qiu and Benbasat, 2009). Unlike immediate reactions, consumer resonance builds gradually, requiring time to deepen (Kouamé et al., 2022). Consumers may initially harbor reservations toward AI virtual streamers; however, as interactions increase, they may develop a sense of resonance that strengthens over time. Information processing theory outlines the cognitive journey from information reception to utilization, involving stages such as encoding, storage, and retrieval (Klahr and Wallace, 2022). Information processing theory provides insights into how disfluency can be integrated into consumers' experiences with AI virtual streamers, potentially intensifying aversion. Following the immediate perceptions of unwarm and incompetent, information

processing theory suggests that consumers engage in more profound cognitive processing, leading to more nuanced cognitive and emotional responses, including disfluency (Li et al., 2022). Therefore, consumer resonance and disfluency, developed at a deeper level of processing, are also integral components of the beliefs formed in the reflective stage, which in turn shape consumers aversion to AI virtual streamers.

The third component, "consequence," refers to the behavioral response to beliefs. Our study focuses on AI virtual streamers aversion. While consumer aversion is recognized as a critical factor that can negatively impact merchant revenue, there has been limited research on consumer aversion to AI virtual streamers (Cheng et al., 2021). Existing research has extensively explored various forms of aversion, including algorithmic aversion (Mahmud et al., 2023), risk aversion (Hossain et al., 2023), loss aversion (Aggarwal and Zhang, 2006), etc. Algorithm aversion is defined as the distrust and dislike of algorithms exhibited by individuals, even when algorithms may surpass human accuracy in certain cases (Turel and Kalhan, 2023). Research has revealed that some individuals harbor a deep-seated distrust of algorithms, leading to their aversion to algorithms regardless of their effectiveness (Liu et al., 2023). This aversion to algorithms is considered an observable behavior (Mahmud et al., 2023), reflecting individuals' emotions and attitudes when interacting with algorithms. Given the fundamental similarities between AI virtual streamers and algorithms, it is conceivable that the manifestation of algorithmic aversion may be equally relevant to the realm of AI virtual streamers. Therefore, we draw on the research framework of algorithmic aversion and migrate it to the domain of AI virtual streamers to explore AI virtual streamers aversion and its antecedents. Previous studies have identified a diverse range of factors contributing to consumer aversion. For instance, research on AI-customer relationship management applications has identified managerial aversion to AI, often driven by a lack of transparency and understanding of the AI decision-making process (Gaczek et al., 2023). Similarly, in organizational decision-making, value, tradition, and image barriers have been identified as significantly linked to algorithm aversion (Mahmud et al., 2023). Previous studies have primarily concentrated on investigating the direct correlation between perceived barriers and aversion behavior, with limited exploration of the underlying psychological mechanisms that drive consumer aversion. Therefore, this research seeks to delve deeper into the psychological factors that trigger consumer aversion, particularly uncovering the dual-stage and dual-dimensional psychological pathways that lead to AI virtual streamers aversion.

2.3. Self-construal theory

Self-construal pertains to the manner in which individuals interpret and situate themselves within a particular referential framework, thereby influencing their self-perception and consciousness. This concept significantly influences their cognitions, emotions, and behaviors (Cross and Markus, 1991). Markus and Kitayama (2014) categorized self-construal into two types: independent and interdependent. These orientations profoundly impact cognitive processes, emotions, motivations, and behaviors. Self-construal is a significant factor in shaping individuals' perceptions and interactions with products, services, and technologies (Zheng et al., 2023). Despite its importance, existing research has yet to fully clarify the specific psychological and behavioral pathways that lead to consumer aversion among different self-construal groups when confronted with AI virtual streamers, as well as the differentiated management strategies necessary to mitigate this aversion effectively.

3. Research model and hypotheses

3.1. Conceptual model

Interpreting the mechanism through which CASA cues and consumer technology cues in driving consumer aversion behaviors, we have developed a research model, as depicted in Fig. 1. Specifically, CASA cues (i.e., anthropomorphism) and consumer technology cues (i.e., technophobia) are proposed as two dimensions as "antecedent". The reflexive stage (i.e., perceived unwarm and perceived incompetent) and the reflective stage (i.e., consumer resonance and disfluency) are included in "belief." AI virtual streamers aversion is the behavioral "consequence" of the model. By highlighting the dynamic relationship between the reflexive stage and the reflective stage, a theoretical explanation is developed for the mediating role of the reflective stage in the pathways through which the reflexive stage influences consumer behavioral consequences. In addition, self-construal is considered a moderator. To address potential alternative factors, we control gender, age, education level, and personal income per month, typically incorporated in studies on consumer behaviors (Xiao et al., 2022).

3.2. The effect of antecedents on the reflexive stage

Technophobia is defined as a complex emotional response to advanced technologies and applications, such as aversive, discomfort, and anxiety, which can be regarded as a comparatively stable personality trait (Khasawneh, 2018; Osiceanu, 2015). Studies have indicated that individuals with high technophobia were prone to exhibit apprehension, fear, and avoidance when engaging with novel technological applications, such as virtual influencers (Deng and Jiang, 2023). Technophobic individuals may feel insecure and unapproachable to advanced technologies (Nimrod, 2018), enhancing the perceived unwarm of negative stereotypes of these entities. Additionally, individuals with high technophobia may perceive situations as unfavorable or threatening, lessening the chance of digital inclusion (Di Giacomo et al., 2019), thus holding negative stereotypes of perceived incompetent in the service efficacy of AI virtual streamers. Given the preceding discussion, the following two hypotheses are put forth:

H1a: Technophobia is positively associated with perceived unwarm.

H1b: Technophobia is positively associated with perceived incompetent.

Anthropomorphism is defined as the inclination of individuals to ascribe human-like traits, emotions, and motivations to non-human objects (Salles et al., 2020). Studies have shown that people frequently attribute human-like characteristics to objects, especially digital ones such as chatbots and artificial intelligence, where anthropomorphizing is comparatively straightforward (Novak and Hoffman, 2019). According to the CASA paradigm, people tend to interact socially with AI entities with human-like characteristics, treating them as human (Nass and Moon, 2000). This paradigm implies that the two dimensions of the stereotype content model can be applied to assess the support for non-human agents. Researchers in the field of marketing have presented abundant empirical evidence for the effects of anthropomorphism on the two dimensions of the stereotype content model (Nass et al., 1994; Nass and Moon, 2000). For example, anthropomorphism in advertising hotel brands enhances perceived warmth, leading to consumers' higher visit intentions (Lee and Oh, 2021). Anthropomorphism is an essential source for consumers to infer the enthusiasm and competence traits of AI virtual streamers (El Hedhli et al., 2023; Liu et al., 2021). These results align with ours, and we argue that the anthropomorphism of AI virtual streamers is likely related to consumers' judgments of their warmth and competence. In particular, attributing human-like emotional and social qualities to AI virtual streamers may encourage greater interaction willingness and foster positive emotional responses and competency evaluations. In essence, anthropomorphism is hypothesized to concurrently mitigate the perceived unwarm and perceived incompetent of AI virtual streamers. In light of the preceding discussion, we hypothesize:

H2a: Anthropomorphism is negatively associated with perceived unwarm.

H2b: Anthropomorphism is negatively associated with perceived incompetent.

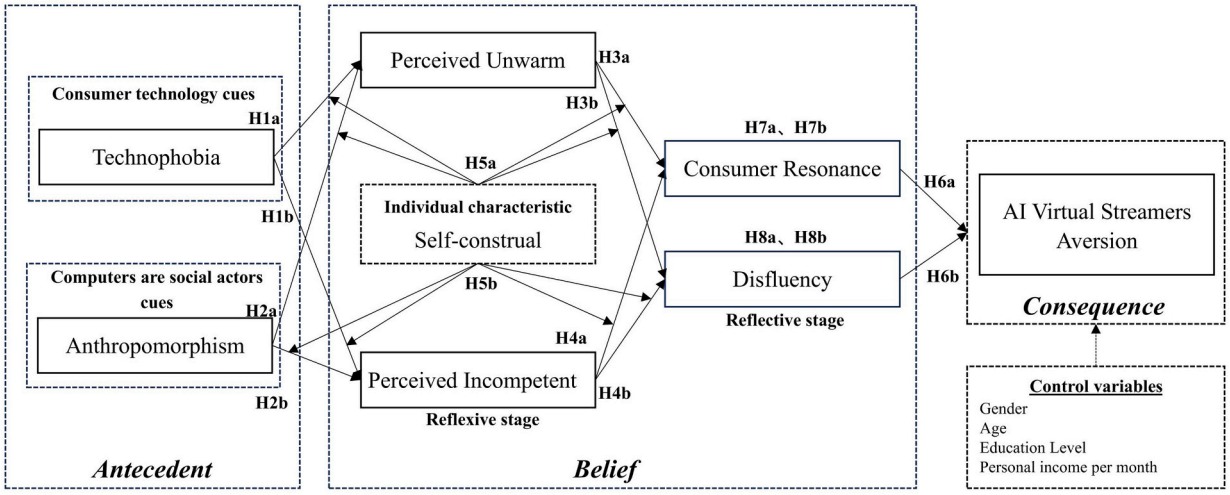


Fig. 1. Research model.

3.3. The effect of the reflexive stage on the reflective stage

Perceived unwarm is defined as consumers' sensation of a lack of human-like care and friendliness (Fiske, 1998). Such perceptions are linked to negative emotions, distrust, and a sense of distance (Fiske et al., 2007). Consumers who perceive AI virtual streamers as unwarm often find it difficult to believe that AI virtual streamers genuinely care about their needs, weakening the emotional bond between consumers and AI virtual streamers (Wan and Jiang, 2023). These emotional barriers of perceived unwarm can also increase the cognitive load when processing information, as consumers need additional mental effort to overcome discomfort (Chi et al., 2023). This extra effort may lead to a sense of emotional and information processing inefficiency for consumers. Consequently, we believe that perceived unwarm toward AI virtual streamers may result in lower levels of identification and consensus, as well as lead to a higher degree of sluggish and stiff consumer experiences. Hence, we develop the following hypotheses for the study:

H3a: Perceived unwarm is negatively associated with consumer resonance.

H3b: Perceived unwarm is positively associated with disfluency.

Perceived incompetent refers to consumers' beliefs that the design quality or technological implementation of AI virtual streamers may be flawed during their interaction process (Fiske, 1998). This perception can lead to lower levels of trust and interaction (Johnson et al., 2008), diminishing the resonance between consumers and AI virtual streamers, thereby reducing support and following. According to cognitive consistency theory, when consumers' experiences do not align with their internal expectations, they are more likely to form negative beliefs, such as disfluency (Kruglanski et al., 2018). Perceived incompetent suggests that consumers may be required to invest additional time and effort in order to assess the accuracy and reliability of the information presented by AI virtual streamers. (Schillaci et al., 2024). This verification process increases the complexity of information processing, adding to the sense of disfluency. In light of these considerations, we propose two hypotheses:

H4a: Perceived incompetent is negatively associated with consumer resonance.

H4b: Perceived incompetent is positively associated with disfluency.

3.4. Moderating role of self-construal

Individuals with independent self-construal perceive themselves as autonomous entities, prioritizing their personal thoughts and distinctive inner needs. Such inward-focused information processors place greater emphasis on the capabilities of AI virtual streamers. Conversely, individuals with interdependent self-construal emotionally perceive themselves as integral to a broader community, valuing social connections and exhibiting greater empathy than their independent counterparts (Jiang et al., 2014). Empirical studies have confirmed the distinctions between the two and their impacts. For example, individuals with an interdependent self-construal give greater importance to emotional communication, while those with an independent self-construal focus more on utilitarian benefits (Aaker and Lee, 2001). Therefore, independent individuals are more sensitive to perceived incompetent, whereas interdependent individuals are more sensitive to perceived unwarm. In other words, independent consumers exhibit aversion to AI virtual streamers due to perceived incompetent, while interdependent consumers are averse to AI virtual streamers more due to perceived unwarm.

H5a: For individuals with interdependent self-construal, cues (i.e., technophobia and anthropomorphism) have a stronger influence on the reflective stage (i.e., consumer resonance and disfluency) by affecting perceived unwarm.

H5b: For individuals with independent self-construal, cues (i.e., technophobia and anthropomorphism) have a stronger influence on the reflective stage (i.e., consumer resonance and disfluency) by affecting perceived incompetent.

3.5. The effect of the reflective stage on consequences

Consumer resonance is characterized as the deep emotional and psychological bond formed between consumers and products or services, manifesting in a sense of harmony (Huang et al., 2014). Studies have shown that resonance with a brand or product correlates with consumer preference and purchase behavior (Shang et al., 2017). In the luxury goods market, brand resonance is crucial for consumer behavior (Husain et al., 2022). Results on service experiences have also indicated that brand resonance positively influences consumer retention (Jang et al., 2021). If consumers perceive a connection with AI virtual streamers, their acceptance is likely to increase, thereby diminishing aversion to AI virtual streamers. Consequently, consumer resonance is anticipated to have an inverse relationship with consumer aversion behaviors, as consumers tend to translate their beliefs into actions.

Disfluency is characterized by the impediments or challenges that arise during information processing, which can be a result of how information is presented, comprehension issues, or a clash with personal experiences (Lickley, 2015). In previous studies, the negative effects of disfluency on target outcomes have been well established (Lee and Labroo, 2004; Novemsky et al., 2007). Zhang et al. (2023) observed an

inverse correlation between disfluency and consumer satisfaction with paid knowledge products. Similarly, Wu et al. (2020) demonstrated that information disfluency leads to lower evaluations by consumers when assessing products, and they are prepared to offer a lower price for the products. Therefore, disfluency may consciously prompt consumers to disregard the offerings recommended by AI virtual streamers. The objective of this study is to examine the relationship between consumer resonance and AI virtual streamers aversion, as well as disfluency and AI virtual streamers aversion, leading to the formulation of the following hypotheses:

H6a: Consumer resonance is negatively associated with AI virtual streamers aversion.

H6b: Disfluency is positively associated with AI virtual streamers aversion.

3.6. Mediating role of consumer resonance

The mediating role of consumer resonance in psychological processes has been empirically validated (Huang et al., 2014; Lithopoulos et al., 2021). For instance, in the realm of social media services, consumer resonance has been identified as a mediator between various types of satisfaction and purchase intentions (Shang et al., 2017). In philanthropic marketing, resonance is considered an essential mediator in driving the impact of philanthropic marketing on consumer satisfaction (Tsui-Hsu Tsai et al., 2014). Expanding upon these insights, we suggest that consumer resonance mediates the relationship between the reflexive stage (i.e., perceived unwarm and perceived incompetent) and AI virtual streamers aversion. The negative stereotypes that consumers hold about AI virtual streamers diminish consumer resonance, thereby fostering consumer aversion behaviors. Consequently, we offer the following two hypotheses:

H7a: Consumer resonance mediates the relationship between perceived unwarm and AI virtual streamers aversion. Specifically, perceived unwarm will decrease consumer resonance, thereby weakening AI virtual streamers aversion.

H7b: Consumer resonance mediates the relationship between perceived incompetent and AI virtual streamers aversion. Specifically, perceived incompetent will decrease consumer resonance, thereby weakening AI virtual streamers aversion.

3.7. Mediating role of disfluency

The mediating role of disfluency has been infrequently explored within the marketing literature. For example, Jaud and Melnyk (2020) demonstrated in their study on wine labels that emotional fluency, rather than cognitive fluency, mediated the relationship between wine labels and purchase intentions. In the context of online reviews, Wu et al. (2022) provided evidence that disfluency mediates the relationship between the multiplicity of emoticons and the effectiveness of comments. In fact, negative stereotypes can influence consumers' initial attitudes of experiencing disfluency, which may manifest as difficulties in technology use, challenges in comprehension, or emotional dissatisfaction. Such disfluency experiences may trigger negative behavioral responses. Therefore, this study proposes the following two hypotheses:

H8a: Disfluency mediates the relationship between perceived unwarm and AI virtual streamers aversion. Specifically, perceived unwarm will enhance disfluency, thereby strengthening AI virtual streamers aversion.

H8b: Disfluency mediates the relationship between perceived incompetent and AI virtual streamers aversion. Specifically, perceived incompetent will enhance disfluency, thereby strengthening AI virtual streamers aversion.

4. Methodology

4.1. Questionnaire development and measures

This research has developed a survey approach to explore research questions. Constructs within the research model were assessed utilizing items sourced from previously validated instruments, which were subsequently adapted to align with the specific context of AI virtual streamers. A seven-point Likert scale was utilized to evaluate the level of agreement for each question item, with response options spanning from 1 (strongly disagree) to 7 (strongly agree). The initial English-language scales were translated into Chinese using the established back-translation technique to ensure the precision and quality of the translations. Subsequently, a two-phase pretesting process was implemented to validate the content of the survey instruments. First, the draft questionnaire has been sent to three academics and two practitioners in the fields of live-streaming commerce and information system design. Several items were slightly reworded based on their feedback to eliminate ambiguity, logical consistency, and contextual appropriateness. Subsequently, a pretest with 40 questionnaires was conducted, showing that the factor loading of an item measuring anthropomorphism was below the acceptable threshold of 0.70, resulting in its exclusion from further analysis. After the deletion, the factor loading of all remaining items was above 0.70, indicating the reliability of the scale. The finalized constructs and corresponding items are presented in Table 1. In line with earlier research methods (Escalas and Bettman, 2005; Chen et al., 2024a, 2024b), a self-construal score was calculated by taking the average interdependent self-construal score from the average independent self-construal score. A higher resulting value indicated a greater inclination toward independence relative to interdependence. Participants who had a self-construal score of 0 were excluded from the dataset.

4.2. Control variables

We incorporated gender, age, education level, and personal income per month as the control variables to mitigate the influence of specific individual characteristics. These demographics were selected as control variables in information systems research due to their frequent recommendation in the literature and their potential impact on an individual's psychological state (Yang et al., 2023).

4.3. Data collection

The empirical analysis was conducted using data collected via an online questionnaire. The first part of the online survey briefly introduces the research team. In the second part, we explicitly emphasized that eligibility for the survey was limited to individuals who understand the concepts and procedures related to AI virtual streamers. In the third section, participants were asked to provide certain demographic information. Finally, we introduced our research and all the constructs to the participants and ensured confidentiality and anonymity.

Two methods were used to determine the sample size for this study. First, we evaluated the sample size requirements for PLS analysis of the proposed model. According to the rule of thumb, which suggests a minimum of 10 cases per indicator (Chin, 1998), a minimum sample size of 40 was necessary, given that the model included up to four indicators. Second, sample size calculations using the priori method are regarded as the optimal approach for determining sample size and efficacy before designing and planning a study. This method effectively reduces the risk of Type I and Type II errors when testing hypotheses (Kang, 2021). The American Psychological Association (APA) Task Force has endorsed and recommended this strategy to ensure sufficient statistical power and an appropriate sample size for empirical research (Peng et al., 2012). We further employed G*Power, a universal statistical power analysis software, to conduct a more rigorous calculation to determine the minimum

Table 1
Research constructs and measurements.

Principle variables	Measurement item
AI virtual streamers aversion (Turel and Kalhan, 2023; Mahmud et al., 2023)	
AVSA1	I trust human streamers more than AI virtual streamers.
AVSA2	I don't think AI virtual streamers' behavior/recommendations are reliable.
AVSA3	Even though the AI virtual streamers may consistently outperform human streamers, I will still choose to watch human streamers.
Disfluency (Mosteller et al., 2014)	
DISF1	I think AI virtual streamers do not convey enough emotion.
DISF2	I think the AI virtual streamers are confusing.
DISF3	I found the AI virtual streamers presentation challenging to understand.
DISF4	I think communication with AI virtual streamers is disfluent.
Anthropomorphism (Aggarwal and McGill, 2007)	
AN1	AI virtual streamers have a mind.
AN2	AI virtual streamers are lifelike.
AN3	AI virtual streamers are like a person.
Consumer resonance (Shang et al., 2017)	
CR1	I will press the "like" button to approve AI virtual streamers.
CR2	I am willing to comment or send gifts to AI virtual streamers.
CR3	I will share the AI virtual streamers' live studio with my friends.
Perceived incompetent (El Hedhli et al., 2023; Fiske, 2018)	
PINC1	AI virtual streamers lack skill.
PINC2	AI virtual streamers are stupid.
PINC3	AI virtual streamers cannot implement their intention.
PINC4	AI virtual streamers are ineffective.
Perceived unwarm (Fiske, 2018)	
PUNW1	AI virtual streamers are emotionless.
PUNW2	AI virtual streamers are cold-looking.
PUNW3	I cannot feel an affinity toward AI virtual streamers.
PUNW4	AI virtual streamers lack enthusiasm.
Technophobia (Khasawneh, 2018)	
TECH1	The performance of AI virtual streamers is likely to elicit feelings of frustration due to their suboptimal capabilities.
TECH2	I think the AI virtual streamers might not perform well.
TECH3	I think using AI virtual streamers would be risky.
TECH4	The effectiveness and trustworthiness of the AI virtual streamers remain uncertain, and whether it would possess the charm that I anticipate is questionable.
Independent self-construal (Hardin et al., 2004; Cross et al., 2011)	
INDS1	Self-sufficiency is a top priority in my life.
INDS2	I pursue my interests regardless of others' opinions.
INDS3	I take pleasure in distinguishing myself from others in various ways.
Interdependent self-construal (Cross and Markus, 1991; Hardin et al., 2004)	
INTS1	I think that my connections with others are more critical than my achievements.
INTS2	I will continue to participate in a group as long as my contributions are deemed necessary, regardless of my level of satisfaction with the group dynamics.
INTS3	I need to maintain harmony within my group.

required sample size, leveraging its capabilities for a priori analysis (Faul et al., 2009). Based on the parameters suggested by Lee and Chen (2022), including an effect size of 0.15, an α of 0.05, and a power (1-β) of 0.95, a sample size of 129 participants was determined to be sufficient for our analysis.

In this research, we employed a non-probability, purposive sampling method for data collection, a technique frequently used to select samples that meet specific criteria or exhibit a common set of traits (Wang and Huang, 2023). The study's target population comprised individuals who possess knowledge of the concepts and procedures associated with AI virtual streamers. The data collection was implemented in July 2024, during the "Taobao Maker Faire" commerce shopping festival. During this period, many merchants on Taobao were at the forefront of integrating and implementing AI virtual streamers in the live-streaming studios (iiMedia, 2024), offering a unique opportunity to gather data on this emerging trend. The questionnaire lasted two weeks through "Questionnaire Star (https://www.wjx.cn)," one of the prominent survey organizations in China. The research employed a snowball sampling approach, beginning with a diverse initial base and expanding through various online platforms, including Xiaohongshu, WeChat, Weibo, Douyin, and professional forums such as China Software Developer Network and Zhihu, to ensure a broader and more representative sample. To enhance the quality of the survey, an attention check item was incorporated into the questionnaire. This item, phrased as "Here is the attention check; please select option B and option C," was strategically placed to ensure that respondents were paying attention and not merely skimming through the questions. Furthermore, following the principle of the online survey, we conducted a thorough verification of participant identification numbers to ensure that each individual was permitted to complete the questionnaire only once. A total of 600 responses were gathered, and attention-check questions were incorporated to uphold the integrity of the collected data. Hence, after carefully checking, 402 samples were used for the statistical analysis. The characteristics of the participants' demographics are presented in Table 2, aligning closely with the demographic profiles of live-streaming commerce users as reported by iiMedia Report (iiMedia., 2023). The report indicates that

Table 2
Demographic analysis ($N = 402$).

	Demographic categories	Levels	Frequency	Percentage
Independent Self-construal (Total: 208)	Gender	Male	78	37.50 %
		Female	130	62.50 %
	Age	<20	32	15.38 %
		20–29	70	33.66 %
		30–39	66	31.73 %
		40–49	21	10.10 %
		>49	19	9.13 %
	Education Level	Junior high school or less	33	15.87 %
		High school	79	37.98 %
		Bachelor's degree	68	32.69 %
		Master's degree or higher	28	13.46 %
	Personal income per month	None	2	0.01 %
		< CNY 2500	48	23.08 %
		CNY 2500–4999	34	16.35 %
		CNY 5000–7499	52	25.00 %
		CNY 7500–9999	43	20.68 %
		≥ CNY10000	29	13.94 %
Interdependent Self-construal (Total: 194)	Gender	Male	77	39.69 %
		Female	117	60.31 %
	Age	<20	28	14.43 %
		20–29	38	19.59 %
		30–39	76	39.18 %
		40–49	28	14.43 %
		>49	24	12.37 %
	Education Level	Junior high school or less	18	9.28 %
		High school	70	36.08 %
		Bachelor's degree	70	36.08 %
		Master's degree or higher	36	18.56 %
	Personal income per month	None	11	5.67 %
		< CNY 2500	38	19.59 %
		CNY 2500–4999	40	20.62 %
		CNY 5000–7499	48	24.74 %
		CNY 7500–9999	39	20.10 %
		≥ CNY10000	18	9.28 %

females constitute 61.0 % of the population, a figure that is closely mirrored in our sample, which comprises 62.5 % females. Additionally, the 20–29 age group is identified as the largest demographic, accounting for 32.5 % of respondents. Our survey similarly shows that this age group is the most represented, comprising 33.66 %. No significant statistical differences were observed in demographic characteristics variables between the first 80 and the last 80 participants.

5. Data analysis

SmartPLS4 analyzed the data using the PLS-SEM method, which encompassed both measurement validation and structural model evaluation. PLS-SEM is a widely used and appropriate method focusing on the interaction between theoretical tests and predictions (Hair et al., 2011). It has advantages in analyzing small samples and non-normally distributed data.

5.1. Measurement validation

The evaluation of the constructs' measurement model involved an examination of their reliabilities, convergent validity, and discriminant validity. First, construct reliability was evaluated using Cronbach's alpha (α) and composite reliability (CR). As illustrated in Table 3, the values of α and CR surpassed the recommended threshold of 0.7 (Fornell and Larcker, 1981), indicating a robust level of reliability within the measurement model.

Convergent validity is generally evaluated by calculating average variance extracted (AVE) values, alongside verifying that the item loadings associated with their respective constructs are adequately elevated (Fornell and Larcker, 1981). In this study, the AVE values exceeding 0.5, as presented in Table 3, indicate convergent solid validity of the scales. Additionally, Table 4 provides evidence that the item loadings associated with their respective constructs exceeded the threshold of 0.7, thereby reaffirming the sufficient convergent validity of our dataset. To assess discriminant validity, three distinct methodologies were employed. Firstly, discriminant validity was evaluated by comparing the item loadings on their respective constructs with cross-loadings on other variables (Chin, 1998). Table 4 illustrates that items exhibit high loadings on their associated constructs and low cross-loadings on other variables, thereby indicating discriminant validity. Secondly, discriminant validity was assessed by comparing the square root of the AVE of each construct with the correlation coefficients related to that construct, as suggested by Fornell and Larcker (1981). As shown in Table 3, the correlations with other constructs were found to be less than the square root of the AVE for each respective construct, indicating strong discriminant validity among the constructs under study. Thirdly, discriminant validity is substantiated by the fact that all Heterotrait-Monotrait (HTMT) latent variable pairs are smaller than 0.85, as shown in Table 5, as recommended by Henseler et al. (2015).

5.2. Common method bias and multicollinearity

Common method bias (CMB) refers to a type of bias that arises when data for both dependent and independent variables are collected from a single source. This bias can be detected by analyzing whether a singular factor predominantly accounts for the variation observed among the variables. (Malhotra et al., 2006). This study has taken several approaches to explore the problem of CMB. First, we evaluated CMB by employing Harman's single-factor test, which demonstrated that a single factor explained 33.162 % of the total variance, a value significantly below 50 %. This finding suggests the absence of CMB in our specific context (Podsakoff et al., 2003). Second, the Unmeasured Latent Method construct technology was put into practice as recommended by Liang et al. (2007). The findings indicated that the average variance attributed to the latent construct (0.735) substantially surpassed that attributed to the common method factor (0.003). The proportion of variance attributable to the substance relative to the variance due to methodology is approximately 245 to 1. Given the substantial ratio, concerns regarding CMB were deemed insignificant. Third, we conducted the marker variable method for addressing method variance, following the recommendation of Lindell and Whitney (2001). A four-item scale measuring online learning satisfaction, adapted from Xiao and Li (2021), was selected as a marker variable, as it is theoretically independent of the constructs within the model. We found correlations between online learning satisfaction and other constructs ranging from −0.088 to −0.011 without any significant correlation. The selected marker variable was then tested according to the construct level correction method (Chin et al., 2013), and the discrepancies in path coefficients and R^2 values between the models that included marketed variables and those that did not were found to be <1 %, suggesting that the CMB does not affect the study results. Fourth, this research investigated the potential presence of multicollinearity by evaluating the variance inflation factors (VIF). As presented in Table 4, all VIF values were found to be below 3.30, suggesting that there are no significant concerns regarding multicollinearity (Kock, 2015). These results align with the perspective proposed by Segijn et al. (2017) that VIF values under 5.00 indicate the absence of CMB. Last but not least, the majority of our pathways involve moderating effects, and existing research indicates that the measurement of these effects is unlikely to be substantially influenced by CMB (Srinivasan and Swink, 2018).

5.3. Structural model analysis

A bootstrapping procedure was performed to obtain the inferential statistics (Hair et al., 2017). The results of the hypothesis testing are displayed in Table 6, including the explained variances and the path coefficients. The results show that technophobia significantly positively affects perceived unwarm and perceived incompetent ($\beta = 0.161$, $t = 2.937$; $\beta = 0.166$, $t = 3.119$), supporting H1a and H1b. Anthropomorphism significantly negatively affects perceived unwarm and

Table 3
Reliability and convergent validity.

Construct	α	CR	AVE	AN	AVSA	CR	DISF	INDS	INTS	PINC	PUNW	TECH
AN	0.817	0.891	0.731	0.855								
AVSA	0.855	0.912	0.776	−0.411	0.881							
CR	0.828	0.897	0.744	0.42	−0.412	0.862						
DISF	0.875	0.914	0.727	−0.378	0.421	−0.364	0.853					
INDS	0.837	0.901	0.753	0.328	−0.406	0.362	−0.384	0.868				
INTS	0.857	0.913	0.777	0.422	−0.445	0.394	−0.385	0.422	0.882			
PINC	0.868	0.91	0.716	−0.393	0.397	−0.389	0.384	−0.419	−0.385	0.846		
PUNW	0.859	0.905	0.703	−0.385	0.384	−0.383	0.321	−0.342	−0.382	0.350	0.839	
TECH	0.870	0.911	0.719	−0.459	0.388	−0.397	0.421	−0.383	−0.442	0.413	0.388	0.848

Note: AN = Anthropomorphism, AVSA = AI Virtual Streamers Aversion, CR = Consumer Resonance, DISF = Disfluency, INDS = Independent Self-construal, INTS = Interdependent Self-construal, PINC = Perceived Incompetent, PUNW = Perceived Unwarm, TECH = Technophobia. The diagonal entries represent the square root of the Average Variance Extracted (AVE).

Table 4
Assessment of measurement model.

Construct	Items	Loadings and cross-loading									VIF
		AN	AVSA	CR	DISF	INDS	INTS	PINC	PUNW	TECH	
AN	AN1	**0.867**	−0.343	0.368	−0.307	0.270	0.383	−0.312	−0.356	−0.429	1.961
	AN2	**0.851**	−0.392	0.378	−0.347	0.311	0.363	−0.383	−0.341	−0.397	1.650
	AN3	**0.848**	−0.311	0.326	−0.314	0.254	0.335	−0.307	−0.285	−0.345	1.927
AVSA	AVSA1	−0.389	**0.882**	−0.378	0.376	−0.353	−0.405	0.348	0.334	0.361	2.097
	AVSA2	−0.346	**0.885**	−0.358	0.390	−0.343	−0.351	0.356	0.308	0.340	2.148
	AVSA3	−0.350	**0.875**	−0.350	0.344	−0.379	−0.422	0.343	0.375	0.322	2.133
CR	CR1	0.357	−0.315	**0.853**	−0.319	0.318	0.335	−0.313	−0.311	−0.335	1.920
	CR2	0.354	−0.353	**0.872**	−0.287	0.293	0.337	−0.357	−0.347	−0.322	1.940
	CR3	0.374	−0.392	**0.862**	−0.337	0.325	0.348	−0.335	−0.332	−0.370	1.812
DISF	DISF1	−0.336	0.380	−0.374	**0.860**	−0.326	−0.317	0.360	0.295	0.359	2.149
	DISF2	−0.341	0.318	−0.285	**0.838**	−0.335	−0.366	0.360	0.294	0.385	2.027
	DISF3	−0.273	0.358	−0.301	**0.848**	−0.303	−0.307	0.296	0.236	0.331	2.194
	DISF4	−0.336	0.377	−0.276	**0.863**	−0.345	−0.324	0.290	0.268	0.361	2.316
INDS	INDS1	0.315	−0.344	0.304	−0.351	**0.865**	0.379	−0.350	−0.293	−0.326	1.978
	INDS2	0.316	−0.347	0.344	−0.349	**0.883**	0.369	−0.406	−0.310	−0.357	1.918
	INDS3	0.213	−0.370	0.289	−0.296	**0.855**	0.351	−0.326	−0.287	−0.310	1.965
INTS	INTS1	0.362	−0.402	0.363	−0.337	0.337	**0.877**	−0.332	−0.345	−0.389	2.043
	INTS2	0.406	−0.418	0.329	−0.362	0.381	**0.889**	−0.339	−0.334	−0.424	2.264
	INTS3	0.350	−0.356	0.351	−0.319	0.400	**0.879**	−0.348	−0.331	−0.356	2.146
PINC	PINC1	−0.338	0.351	−0.326	0.338	−0.351	−0.322	**0.837**	0.272	0.329	2.000
	PINC2	−0.338	0.339	−0.331	0.310	−0.337	−0.311	**0.850**	0.302	0.352	2.161
	PINC3	−0.302	0.297	−0.293	0.336	−0.351	−0.305	**0.836**	0.283	0.328	2.037
	PINC4	−0.350	0.353	−0.365	0.319	−0.378	−0.362	**0.862**	0.325	0.387	2.222
PUNW	PUNW1	−0.338	0.324	−0.338	0.280	−0.283	−0.314	0.269	**0.844**	0.342	2.001
	PUNW2	−0.299	0.348	−0.300	0.260	−0.269	−0.309	0.320	**0.823**	0.304	1.934
	PUNW3	−0.323	0.297	−0.342	0.301	−0.324	−0.323	0.284	**0.846**	0.345	1.998
	PUNW4	−0.331	0.322	−0.302	0.233	−0.270	−0.337	0.303	**0.841**	0.305	2.075
TECH	TECH1	−0.367	0.323	−0.321	0.315	−0.300	−0.375	0.343	0.335	**0.839**	2.012
	TECH2	−0.392	0.322	−0.349	0.326	−0.284	−0.359	0.357	0.314	**0.848**	2.100
	TECH3	−0.398	0.313	−0.341	0.382	−0.372	−0.353	0.364	0.287	**0.849**	2.148
	TECH4	−0.399	0.356	−0.337	0.405	−0.344	−0.409	0.338	0.375	**0.857**	2.147

Note: AN = Anthropomorphism, AVSA = AI Virtual Streamers Aversion, CR = Consumer Resonance, DISF = Disfluency, INDS = Independent Self-construal, INTS = Interdependent Self-construal, PINC = Perceived Incompetent, PUNW = Perceived Unwarm, TECH = Technophobia.

Table 5
Heterotrait-monotrait (HTMT) ratios.

Construct	AN	AVSA	CR	DISF	INDS	INTS	PINC	PUNW	TECH
AN									
AVSA	0.487								
CR	0.507	0.486							
DISF	0.445	0.485	0.426						
INDS	0.389	0.482	0.432	0.447					
INTS	0.503	0.520	0.468	0.445	0.499				
PINC	0.463	0.459	0.457	0.440	0.487	0.446			
PUNW	0.456	0.450	0.452	0.368	0.401	0.445	0.405		
TECH	0.541	0.448	0.468	0.482	0.447	0.511	0.475	0.446	

Note: AN = Anthropomorphism, AVSA = AI Virtual Streamers Aversion, CR = Consumer Resonance, DISF = Disfluency, INDS = Independent Self-construal, INTS = Interdependent Self-construal, PINC = Perceived Incompetent, PUNW = Perceived Unwarm, TECH = Technophobia.

Table 6
Results of the direct effects.

Path	Hypothesis	Effect	T-statistic	*P*-values	Result
TECH→PUNW	H1a	0.161	2.937	0.003	Supported
TECH→PINC	H1b	0.166	3.119	0.002	Supported
AN→PUNW	H2a	−0.160	3.021	0.003	Supported
AN→PINC	H2b	−0.171	3.437	0.001	Supported
PUNW→CR	H3a	−0.282	6.257	0.000	Supported
PUNW→DISF	H3b	0.213	4.527	0.000	Supported
PINC→CR	H4a	−0.291	6.396	0.000	Supported
PINC→DISF	H4b	0.310	6.745	0.000	Supported
CR → AVSA	H6a	−0.298	6.949	0.000	Supported
DISF→AVSA	H6b	0.312	7.013	0.000	Supported

Note: AN = Anthropomorphism, TECH = Technophobia, AVSR = AI Virtual Streamers Aversion, CR = Consumer Resonance, DISF = Disfluency, PUNW = Perceived Unwarm, INDS = Independent Self-construal, PINC = Perceived Incompetent.

incompetent ($\beta = -0.160$, $t = 3.021$; $\beta = -0.171$, $t = 3.437$), supporting H2a and H2b. Perceived unwarm significantly affects consumer resonance and disfluency ($\beta = -0.282$, $t = 6.257$; $\beta = 0.213$, $t = 4.527$), supporting H3a and H3b. Similarly, perceived incompetent significantly affects consumer resonance and disfluency ($\beta = -0.291$, $t = 6.396$; $\beta = 0.310$, $t = 6.745$), supporting H4a and H4b. Consumer resonance negatively affects AI virtual streamers aversion ($\beta = -0.298$, $t = 6.949$), supporting H6a. Furthermore, disfluency is positively and significantly associated with AI virtual streamers aversion ($\beta = 0.312$, $t = 7.013$). Thus, H6b is supported.

Moreover, the collective influence of all variables accounts for 54.5 % of the variance observed in AI virtual streamer aversion. For perceived unwarm, perceived incompetent, consumer resonance, and disfluency, the values were 40.50 %, 38.80 %, 27.10 %, and 28.80 %, respectively (see Fig. 2).

We conducted additional examinations to assess the impact of the control variables within the model. There is no statistically significant

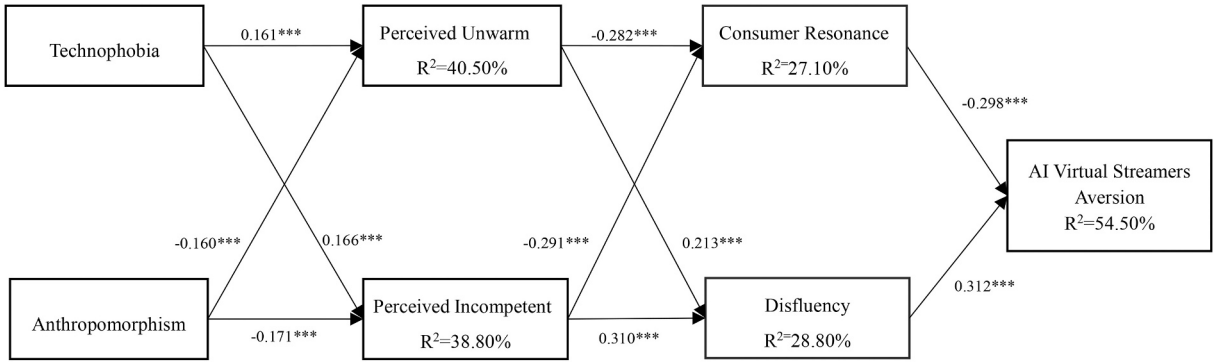


Fig. 2. Results of the structural model.

correlation observed between gender (β = 0.005, *t* = 1.264), age (β = 0.033, *t* = 0.159), education level (β = 0.019, *t* = 0.104), personal income per month (β = 0.013, *t* = 0.802) and AI virtual streamers aversion.

5.4. Mediation testing

The study employed bootstrapping methodology to evaluate the mediating effect of perceived unwarm, perceived incompetent, consumer resonance, and disfluency using SmartPLS4. A total of 5000 bootstrapped samples were generated for this analysis. The significance of the mediating effect is established when the 95 % confidence interval of the estimated effect does not encompass zero (Preacher and Hayes, 2008), confirming that the mediation effects in this study are statistically significant, as illustrated in Table 7. In addition, Table 7 illustrates both the direct and indirect effects, thereby showcasing the mediation analysis performed in accordance with the methodology established by Zhao et al. (2010), confirming the partial mediating roles of perceived unwarm, perceived incompetent, consumer resonance, and disfluency.

Furthermore, the strength of partial mediation can be assessed by determining the variance accounted for (VAF). According to Hair et al. (2019), a VAF exceeding 0.2 indicates the presence of a mediation effect. If the VAF is between 0.2 and 0.8, the mediation is deemed partial; however, if the VAF surpasses 0.8, it is considered complete mediation. As shown in Table 7, anthropomorphism partially mediates consumer resonance and disfluency through perceived unwarm, with VAF values of 33.184 % and 29.167 %, respectively. Anthropomorphism also has partially mediating effects on consumer resonance and disfluency through perceived incompetent, with VAF values of 33.482 % and 37.037 %, respectively. Moreover, technophobia partially mediates consumer resonance through perceived unwarm and perceived incompetent, with VAF values of 31.780 % and 34.818 %, respectively. Technophobia also partially mediates disfluency through perceived unwarm and perceived incompetent, with VAF values of 27.805 % and 38.075 %, respectively. Finally, perceived unwarm and perceived incompetent decreased consumer resonance, which led to AI virtual streamers aversion, with VAF values of 35.745 % and 32.222 %. Moreover, perceived unwarm and perceived incompetent enhanced disfluency, which led to AI virtual streamers aversion, with VAF values of 30.734 % and 34.643 %. These results supported H7a, H7b, H8a, and H8b on the partial mediation roles of consumer resonance and disfluency.

5.5. Multigroup analysis (MGA)

Multigroup analysis is employed to compare differences in behavior, attitudes, or preferences among various groups (Sarstedt et al., 2011). Through MGA, researchers can segment data into multiple groups based on different levels of the moderators. This approach enables the estimation of model parameters for each group independently, thereby elucidating the impact of the moderators on the relationship between independent and dependent variables. (Matthews, 2017; Rubio-Andrés et al., 2023; Lu and Xu, 2024). SmartPLS4 offers convenient tools that facilitate direct MGA analysis, enabling researchers to process and compare data across different groups efficiently (Cheah et al., 2023). We initially confirmed measurement invariance through a three-stage process in MICOM before proceeding with the MGA analysis, which is essential for ensuring valid comparisons and insights into the group-specific variations derived from the PLS-SEM findings (Liengaard, 2024). Table 8 demonstrates that the MICOM process, following Henseler et al. (2016), successfully validated the configurational invariance, compositional invariance, and partial measurement invariances. This validation is essential for ensuring the legitimacy of comparing and interpreting differences between groups within the MGA. Following the

Table 7
Mediating effect results.

Model path	Direct effect	Indirect effect	Total effect	Confidence interval	T-statistic	VAF (%)
TECH→PINC→DISF	0.148***	0.091***	0.239	[0.055, 0.144]	4.078	38.075 %
TECH→PUNW→DISF	0.148***	0.057***	0.205	[0.027, 0.093]	3.330	27.805 %
TECH→PINC→CR	−0.161***	−0.086***	−0.247	[−0.134, −0.050]	4.142	34.818 %
TECH→PUNW→CR	−0.161***	−0.075***	−0.236	[−0.111, −0.043]	4.321	31.780 %
AN→PINC→DISF	−0.136***	−0.080***	−0.216	[−0.123, −0.048]	4.079	37.037 %
AN→PUNW→DISF	−0.136***	−0.056***	−0.192	[−0.092, −0.027]	3.224	29.167 %
AN→PINC→CR	0.149***	0.075***	0.224	[0.042, 0.115]	4.092	33.482 %
AN→PUNW→CR	0.149***	0.074***	0.223	[0.043, 0.113]	3.925	33.184 %
PUNW→CR → AVSA	0.151***	0.084***	0.235	[0.050, 0.124]	4.405	35.745 %
PINC→CR → AVSA	0.183***	0.087***	0.270	[0.050, 0.130]	4.239	32.222 %
PUNW→DISF→AVSA	0.151***	0.067***	0.218	[0.034, 0.107]	3.582	30.734 %
PINC→DISF→AVSA	0.183***	0.097***	0.280	[0.059, 0.142]	4.591	34.643 %

AVSA = AI Virtual Streamers Aversion, CR = Consumer Resonance, DISF = Disfluency, PUNW = Perceived Unwarm, PINC = Perceived Incompetent, AN = Anthropomorphism, TECH = Technophobia.
*** Note: $p < 0.001$.

Table 8
Results of invariance measurement.

Constructs	Step 1: Configurational invariance	Step 2: Compositional invariance		Partial measurement invariance established	Step 3a: equal mean assessment			Step 3b: equal variance assessment			Full measurement invariance established
		Original correlation	5 %		Differences	2.50 %	97.50 %	Differences	2.50 %	97.50 %	
AN	Yes	0.999	0.997	Yes	0.188	−0.202	0.206	0.198	−0.202	0.202	Yes
CR	Yes	0.998	0.997	Yes	0.017	−0.191	0.188	0.094	−0.201	0.201	Yes
DISF	Yes	0.992	0.998	Yes	−0.071	−0.188	0.19	−0.046	−0.201	0.212	Yes
PINC	Yes	0.993	0.998	Yes	−0.063	−0.199	0.188	−0.052	−0.204	0.197	Yes
PUNW	Yes	0.998	0.998	Yes	−0.062	−0.190	0.186	−0.162	−0.202	0.194	Yes
TECH	Yes	0.999	0.992	Yes	−0.097	−0.204	0.185	−0.145	−0.189	0.188	Yes

Note: AN = Anthropomorphism, CR = Consumer Resonance, DISF = Disfluency, PUNW = Perceived Unwarm, PINC = Perceived Incompetent, TECH = Technophobia.

establishment of measurement invariance, MGA was performed to compare the results from independent and interdependent self-construal, as shown in Table 9. The findings suggest that consumers' self-construal serves as a moderator in the relationships among two cues, perceived unwarm and the reflective stage. Specifically, when consumers are interdependent self-construal, two cues have a stronger indirect effect on the reflective stage through perceived unwarm than consumers are independent self-construal. Thus, H5a was supported.

Moreover, consumers' self-construal also moderated the relationships among two cues, perceived incompetent and the reflective stage. Especially when consumers are independent self-construal, two cues have a stronger indirect effect on the reflective stage through perceived incompetence than when consumers are interdependent self-construal, supporting H5b.

6. Discussion

6.1. Discussion of results

Drawing on the ABC framework, this research elucidates the process through which CASA cues (i.e., anthropomorphism) and consumer technology cues (i.e., technophobia) sequentially lead to the reflexive stage (i.e., perceived unwarm and perceived incompetent) and the reflective stage (i.e., consumer resonance and disfluency), culminating in AI virtual streamers aversion. This consequence is further moderated by the individual characteristics in consumers' self-construal.

First, this study examines how anthropomorphism and technophobia influence consumers' negative perceptions of AI virtual streamers. Consistent with the findings of El Hedhli et al. (2023), the study shows a positive correlation between technophobia and negative stereotypes of AI virtual streamers. Specifically, as technophobia increases, so does the likelihood of holding negative stereotypes, likely due to heightened scrutiny of the AI virtual streamers' capabilities, a tendency driven by technophobia (Di Giacomo et al., 2019). On the other hand, anthropomorphism is negatively associated with these stereotypes and may even reduce aversion, aligning with the results of Seymour et al. (2024). We interpret this finding as indicating that anthropomorphism enhances consumers' perception of social presence by providing emotional gratification (Lee et al., 2023), which in turn mitigates the negative stereotypes. Consequently, alleviating technophobia and refining the anthropomorphic cues of AI virtual streamers could be instrumental in fostering favorable consumer perceptions and responses.

Second, this study contributes empirical evidence by elucidating the mediating role of the reflective stage (i.e., consumer resonance and disfluency) in negative stereotypes (i.e., perceived incompetent and perceived unwarm) and AI virtual streamers aversion. The results substantiate the significant impact of the reflective stage on consumer aversion, aligning with prior studies that associate empathy with heightened repurchase intentions (Sharma et al., 2024). However, these findings challenge the assertion made by De Cicco et al. (2024) that the acceptance of virtual influencers is not impeded by a communication style based on emotions. This finding reinforces the notion that enhancing consumer resonance and reducing disfluency can be effective strategies for reducing consumer aversion to AI virtual streamers.

Third, we examine self-construal as a boundary condition of AI virtual streamers aversion. The results indicate that interdependent consumers are averse to AI virtual streamers due to perceived unwarm, reflecting their prioritization of emotional connection and intimacy in services. In contrast, independent consumers are averse to AI virtual streamers due to perceived incompetence, which aligns with the high value placed on efficacy and competency. Our study examines self-construal as a boundary condition for consumer aversion,

Table 9
Multigroup analysis with self-construal as a moderator.

H. No.	Mediating path	Self-construal path coefficient		Confidence interval		Path Coefficient Difference	T-statistic	P-value	Results
		INDS	INTS	INDS	INTS				
H5a	TECH→PUNW→DISF	0.021^{NS}	0.063*	[−0.007,0.061]	[0.019, 0.125]	−0.042*	1.968	0.050	Supported
	TECH→PUNW→CR	0.022^{NS}	0.109**	[−0.005, 0.061]	[0.044,0.188]	−0.059*	2.508	0.013	Supported
	AN→PUNW→DISF	−0.027^{NS}	−0.059*	[−0.068, 0.006]	[−0.119, −0.017]	0.033**	2.856	0.005	Supported
	AN→PUNW→CR	−0.017^{NS}	−0.116**	[−0.054,0.007]	[−0.189, −0.053]	0.099*	2.086	0.038	Supported
H5b	TECH→PINC→DISF	0.115*	0.012^{NS}	[0.042, 0.207]	[−0.017, 0.056]	0.102*	2.192	0.029	Supported
	TECH→PINC→CR	0.030*	0.003^{NS}	[0.012, 0.063]	[−0.001, 0.015]	0.026*	1.794	0.074	Marginally Supported
	AN→PINC→DISF	−0.125*	−0.010^{NS}	[−0.220, −0.070]	[−0.048, 0.009]	−0.115**	2.856	0.005	Supported
	AN→PINC→CR	−0.042*	−0.002^{NS}	[−0.089, −0.018]	[−0.011, 0.011]	−0.040**	3.478	0.001	Supported

Note: AN= Anthropomorphism, TECH= Technophobia, CR= Consumer Resonance, DISF= Disfluency, PUNW = Perceived Unwarm, PINC= Perceived Incompetent, INDS=Independent Self-construal, INTS=Interdependent Self-construal; NS=No significant.

* $p < 0.05$.

** $p < 0.01$.

complementing previous findings that different self-construal led to varying consumer behavior toward technology (Zheng et al., 2023; Chen et al., 2024a, 2024b). Furthermore, our research indicates that self-construal serves as a moderating factor in individual differences, thereby supporting scholarly literature that examines the impact of individual variances on consumer behavior (Mahmud et al., 2023; Liu et al., 2023).

6.2. Theoretical contributions

The theoretical contributions are fourfold. First, this study constructs and validates a research model for consumer aversion to AI virtual streamers, thereby providing a significant enhancement to the current research framework in the field of live-streaming commerce. While previous studies have primarily focused on human streamers (Wang et al., 2024; Zhang et al., 2023), this study enriches the live-streaming commerce literature by shifting the focus to AI virtual streamers. From a research perspective, prior studies have predominantly emphasized positive outcomes like purchase intention and willingness to accept (Sun et al., 2024; Chen et al., 2024a, 2024b). This study addresses a notable gap in existing literature by investigating AI virtual streamers aversion, which has largely overlooked negative consumer responses to AI virtual streamers. Consequently, it expands the theoretical horizons of research on live-streaming commerce and encourages scholars to explore the negative effects of AI virtual streamers on consumer behavior. Furthermore, we have developed a robust theoretical foundation for understanding consumer aversion to AI virtual streamers. By effectively adapting the theory of algorithmic aversion to the context of AI virtual streamers, this cross-domain theoretical integration expands existing knowledge and offers valuable insights for future research.

Second, this research represents a noteworthy advancement by synthesizing technological features and consumer traits within a unified framework. While the role of consumer traits has been widely recognized in emerging technology service research (Chuah et al., 2021; Xiao et al., 2022), existing studies on AI virtual streamers have inadequately addressed the influence of consumer traits, focusing primarily on technological features (Gao et al., 2023; Wang et al., 2024; Chen et al., 2024a, 2024b). This research addresses this gap by incorporating consumer traits (i.e., technophobia) alongside technological features (i.e., anthropomorphism) to investigate AI virtual streamers aversion through the lens of complexity theory. Our research contributes to a deeper comprehension of the origins of AI virtual streamer aversion, indicating that individuals with technophobic tendencies are more prone to such aversion. Conversely, anthropomorphic design elements can help mitigate such aversion. By applying complexity theory, our study significantly boosts its impact in the realm of live-streaming commerce and helps to understand the antecedents to consumer aversion to AI virtual streamers.

Third, this research offers corroborating evidence supporting the compatibility of the temporal model of rejection response with the ABC framework, particularly by integrating reflexive stages and reflective states into the "belief" component of the ABC model, thereby extending its applicability. Previous research on consumer psychological mechanisms has traditionally focused on the reflexive stages (such as empathy) (Xie et al., 2024) or reflective states (such as trust) (Wang et al., 2024). However, this study uniquely investigates the beliefs contributing to AI virtual streamers aversion, considering both the reflexive stages (i.e., perceived incompetent and perceived unwarm) and the reflective stage (i.e., consumer resonance and disfluency). These stages collectively reveal the dynamics and complexity of consumer psychological mechanisms and provide insights for developing short-term and long-term effective management strategies to reduce consumer aversion to AI virtual streamers.

Fourth, our research extends the understanding of AI virtual streamers aversion by considering consumers' self-construal as a moderating factor, providing a novel theoretical perspective on how individual characteristics shape consumers' perceptions and responses to AI virtual streamers. While individual characteristics have been confirmed to influence consumer behavior significantly (Mahmud et al., 2023; Zheng et al., 2023; Deng and Jiang, 2023), there is limited research exploring how these characteristics impact their responses to AI virtual streamers. In this study, we examine whether consumers with different self-construal exhibit distinct reasons for AI virtual streamers aversion. The results show that for independent consumers, perceived incompetent is a stronger mediator between two cues and the reflective stage, whereas for interdependent consumers, perceived unwarm plays a more significant mediating role. Our research enhances the comprehension of the conditions under which two cues influence consumer aversion.

6.3. Managerial implications

Our work provides valuable insights regarding managerial implications. First, designers and developers of AI virtual streamers should prioritize anthropomorphic features. Our findings indicate that enhancing anthropomorphism can effectively counter negative stereotypes, foster consumer resonance, and reduce disfluency, thereby diminishing consumer aversion. To improve the perception of anthropomorphism among consumers, developers should consider adopting a multidimensional approach to enrich anthropomorphic elements. This may involve enhancing visual appeal, optimizing vocal tone, pitch, and speech rate, as well as equipping with logical dialogue and interactive skills. Additionally, addressing issues such as unnatural movements and repetitive language is essential to improve service fluidity.

Second, virtual live-streaming service operators need to explore segmentation methods that actively consider consumers' self-construal. Currently, AI virtual streamers employ a "one-size-fits-all" approach in their interactions with consumers, resulting in standardized responses to consumers with diverse individual characteristics (Chen, 2023). Our empirical evidence indicates that independent consumers may develop aversion due to a perceived unwarm, while interdependent consumers may feel aversion stemming from perceived incompetent. Operators can identify self-construal segments by analyzing consumer data, such as comment patterns, browsing history and interaction content with AI virtual streamers. With this information, they can provide more emotional support for interdependent consumers and focus on delivering practical information and feedback to independents.

Third, our findings can assist live-streaming commerce platforms in predicting consumer reactions when deploying AI virtual streamers. The results emphasize that technophobia can lead to AI virtual streamers aversion, stemming from opposing views on AI technology or a failure to recognize the benefits of AI virtual streamers. In contrast, consumers with low technophobia are more receptive to new services (Filieri et al., 2022). Therefore, platforms should align their marketing strategies with the technological preferences of their target consumers. Additionally, platforms are advised to implement panel surveys to track consumers' judgments of AI virtual streamers and develop optimization measures.

6.4. Limitations and future study

Firstly, considering that our research focused on the Chinese market, where live-streaming commerce has achieved significant prosperity, it may not be wise to apply our findings to different international contexts indiscriminately. Future studies could collect empirical data among consumers in diverse national contexts to determine if our research findings have cross-cultural validity. Secondly, incorporating additional predictors, such as the lack of empathy and information overload, could enhance the predictive efficacy regarding consumer aversion. Diverging from the mediating mechanisms presented in our research, future research could explore the mechanism explanation between variables using alternative theoretical perspectives, such as distrustful beliefs. Moreover, examining moderating variables, including product type and

brand identity, could clarify more precise boundary conditions for this line of inquiry. Thirdly, while snowball sampling is effective for reaching a broader audience, it may introduce biases due to its potential lack of representativeness of the overall population being studied. Future research endeavors may be enhanced by the implementation of a broader range of sampling methodologies. to obtain a more balanced and representative sample. Finally, a cross-sectional survey-based research design limits our ability to determine causality conclusively. Hence, future research could involve multi-wave surveys and longitudinal investigations to track the evolution of the outcomes associated with these hypotheses as they unfold over an extended period.

CRediT authorship contribution statement

Quan Xiao: Writing – review & editing, Methodology, Conceptualization. **Xia Li:** Writing – original draft, Data curation. **Weiling Huang:** Validation. **Xing Zhang:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A
Positioning of this study in existing literature.

Study	Research motivation 1 Dependent variable	Research motivation 2 Independent variable	Research motivation 3	
			Mediator	Moderator
Prior literature	**Positive outcomes:** Such as purchase intention (Hu and Ma, 2023), consumption willingness (Yu et al., 2023), and trust (Wang and Huang, 2023)	**Technical features:** Such as anthropomorphism (Chen et al., 2024a, 2024b), types of virtual streamers (Wan and Jiang, 2023), and coolness (Zhang et al., 2023)	**Consumers' perceptions:** Such as perceived eeriness and perceived affinity (Liu et al., 2024), perceived flexibility (Chang and Wang, 2023)	**Individual difference:** Such as technology readiness (Mahmud et al., 2023), socio-demographic information (Liu et al., 2023)
This study	**Negative outcomes:** AI virtual streamers aversion	**Technical features:** Anthropomorphism **Consumer traits:** Technophobia	**Consumers' perceptions and chain responses:** Perceived unwarm, perceived incompetent, consumer resonance, disfluency	**Individual difference:** Self-construal

Table B
Relevant literature on AI virtual streamers in live-streaming commerce.

Study	Methodology	Independent variable	Underlying mechanism	Boundary condition	Dependent variable
Liu et al. (2021)	Survey	Personal brand essence, personal brand heritage, realistic plot, credible advertising message, customer response expertise, customer response speed	Intimacy, similarity, authenticity, customer response capability	–	Online engagement
Wu et al. (2022)	Experiments	Socialness	Social presence	Communication style	Experiential value
Wan and Jiang (2023)	Interviews and experiments	Virtual vs human influencer	Perceived ease of use, perceived enjoyment, perceived usefulness, perceived dialogue, influencer warmth, perceived social distance, influencer trust	–	Purchase intention
Zhou et al. (2023)	Experiments	Influencer type	Imagery difficulty, sensory capacity	Sensory type	Purchase intention
Hu and Ma (2023)	Experiments	Sensory language	Language expectancy violation	Human-backed or AI-backed virtual streamers	Purchase intention
Gao et al. (2023)	Experiments	Likeability, animacy, responsiveness	Social presence, telepresence	–	Purchase intention
Zhang et al. (2023)	Survey	Coolness, mind perception, congruence	Arousal	Mindset, impulsiveness	Parasocial interaction intention, urge to buy impulsively
Wang et al. (2024)	Experiments	Ability, benevolence, predictability, integrity, social presence, perceived enjoyment, perceived similarity	Trust	–	Purchase intention

(*continued on next page*)

Table B (*continued*)

Study	Methodology	Independent variable	Underlying mechanism	Boundary condition	Dependent variable
Chen (2023)	Literature review and case studies	Virtual anchors present advantages and challenges	–	–	Potential, operational models, profit methods
Yu et al. (2023)	Survey	Physical attractiveness, interactivity, entertainment	–	–	Consumption willingness
Chen et al. (2024a, 2024b)	Experiments	Anthropomorphism	Psychological distance, trust	Product type	Willingness to accept
Xie et al. (2024)	Experiments	Streamer type	Consumer empathy	Brand reputation, streamer influencer	Consumer brand forgiveness
Sun et al. (2024)	Survey	Anthropomorphism, media richness	Psychological distance, customer engagement	–	Purchase intention

Data availability

Data will be made available on request.

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